The Nuance of Autonomy: Female Household Headship and Fertility in Taiwan

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1 Introduction

This study investigates the causal relationship between a woman's intra-household power—proxied by her status as a female household head—and the timing of her first childbirth. Specifically, we ask: **Does being a female household head affect a woman's age at first birth?** Our hypothesis is that greater household autonomy may enable women to delay childbearing in line with their preferences.

This question matters because it speaks to both theoretical and policy-relevant debates. Theoretically, it extends the literature on fertility timing by linking reproductive outcomes to decision-making power within the household, drawing on frameworks such as Kabeer (1999), who conceptualizes women's empowerment through resources, agency, and achievements. Empirically, it addresses the rising concern over delayed fertility in East Asia and the role of autonomy in shaping demographic trends.

To answer this question, we leverage the 2013 Taiwanese Women's Marital and Employment Status Survey, which includes detailed information on household structure, reproductive history, education, and partner characteristics. Our empirical strategy employs the Post-Double Selection (PDS) Lasso procedure for variable selection in a high-dimensional covariate space. We control for both education group and birth cohort fixed effects to account for secular trends and nonlinear educational gradients.

Our findings suggest that women identified as female household heads tend to have their first birth slightly earlier than their non-head counterparts. This result remains robust across model specifications and sample restrictions. Notably, even after limiting the analysis to currently married women, the direction of the effect persists—challenging the assumption that female headship necessarily reflects autonomy and delayed fertility. Instead, it may capture underlying structural vulnerabilities—such as economic strain, informal separation, or partial spousal absence—that shape both household roles and fertility timing in this context.

2 Literature Review

To understand how household decision-making dynamics shape fertility outcomes, Kabeer (1999) offers a foundational conceptual framework of women's empowerment, emphasizing three components: resources, agency, and achievements. Among them, *agency*—the ability to define one's goals and act upon them—is particularly relevant to reproductive behavior.

In this view, fertility timing is not merely a function of individual preferences or socioeconomic status, but also of the extent to which women can exercise control over household decisions. Even when women have

access to material or social resources, a lack of agency may inhibit the realization of delayed childbearing goals.

Our study builds on this perspective by treating female headship as a proxy for intrahousehold autonomy. If women who are recognized as household heads enjoy greater decision-making authority, this may translate into distinctive fertility patterns, such as delayed age at first birth. Conversely, in patriarchal settings, female headship might not entail real power, raising the need for empirical scrutiny.

3 Data and Samples

Our sample was constructed from the **2013 nationwide Women Marital and Employment Status Questionnaire** data, collected between August 1st and August 31st, 2013. The initial dataset comprised **28,450 samples** of women aged between 15 and 64. After variable construction and exclusion procedures detailed below, this full sample was retained for analysis.

We created the key variables for our analysis:

- `female_head`**:** This binary variable serves as our treatment variable and a proxy for fertility autonomy. It was derived from the categorical variable `a1: relationship_to_head`. Specifically, `relationship_to_head == 2` (representing "relationship_to_head") was recoded as `female_head = 1` (female household head), while `relationship_to_head == 1` was recoded as `female_head = 0` (male household head). Other family members as household heads were excluded to maintain a direct focus on female fertility autonomy.
- `age_at_first_birth`**:** This is our primary outcome variable, measuring the core indicator of fertility timing. It was calculated by subtracting the respondent's estimated birth year from the first birth year. The respondent's birth year was derived as `respondent_birth_yyyy = 1911 + 102 - age`. The first birth year was obtained by taking the floor of `df["first_birth_yyyymm"]` divided by 100.

The final sample size for the analysis, after these cleansing and variable construction steps, is 13,856, drawn from the initial survey data collected in 2013.

Since the data were collected through a **cross-sectional survey** comprising **176** structured questions, certain limitations are inherent. To mitigate these and ensure analytical validity, several measures were implemented. We **excluded all post-treatment variables** that reflect fertility outcomes, employment status, or childcare arrangements occurring after the first birth, in order to avoid future-leakage and maintain correct causal ordering. **Covariates with high multicollinearity** were filtered using a variance inflation factor (VIF) threshold of 10, ensuring model stability. To reduce structural heterogeneity, the sample was **restricted to women who were either household heads or spouses of household heads**.

To address potential confounding due to societal and policy shifts across generations, we **categorized respondents into four birth cohorts** (1940–54, 1955–69, 1970–84, and 1985–94) and included **cohort fixed effects** in all model specifications. We also controlled for **five educational groups**—namely, Junior College, College, Graduate, and PhD—by introducing dummy variables, using **High School or Below** as the reference category. This set of controls ensures that our estimated effects are not confounded by systematic differences across generations or educational attainment levels.

Despite these design choices, some limitations remain. **Causal inference is constrained by the cross-sectional nature** of the data, as temporal ordering between female headship and fertility timing cannot be fully established. **Reverse causality may still exist**, since early childbearing could lead to household headship through divorce or widowhood. Moreover, as the survey reflects the social and institutional context of Taiwan in 2013, **external generalizability to other countries or periods is limited**.

- **Exclusion of Invalid Covariates:** We excluded variables that were either identifiers, geographically irrelevant, or temporally mismatched, as listed below:
  - **Identifying variables:** `sample_id`, `household_id`, `person_id`, `partner_id`, `spouse_id`
  - **Administrative and regional identifiers:** `area_code`, `county_code`, `township_code`, which indicate location but do not contribute analytically to individual-level fertility outcomes.
  - **Temporally mismatched variables:** `partner_retired_status`, which may not accurately reflect the partner's status at the time of the respondent's first birth
- **Exclusion of Future-Leak Covariates.** To prevent post-outcome bias and ensure proper causal inference, we excluded variables that contain information reflecting events, decisions, or statuses that occurred after the respondent's first birth. These variables fall into the following categories:
  - **Fertility outcomes:** `births_boy`, `second_birth_yyyymm`, `no_more_children`...
  - **Childcare arrangements:** `care_baby_under3_who`, `care_baby_3to6_time`, `care_hours_kids`...
  - **Post-birth employment behaviors:** `worked_last_week`, `work_hours_main`, `reason_not_working`...
  - **Return-to-work after birth:** `backtowork_afterbirth_age`, `backtowork_afterbirth_title`, `maternity_quit_return_status`...
  - **Future employment preferences:** `changejob_afterbirth_title`...

  These covariates were strictly excluded to prevent data leakage that could otherwise inflate the model's explanatory power through access to information not available at the time of treatment assignment.
- **Handling Multicollinearity:** Prior to LASSO selection, a multicollinearity test is performed, and covariates with a Variance Inflation Factor (VIF) greater than 10 are filtered out to ensure the stability and interpretability of coefficient estimates.
  - **Partner demographics:** `partner_gender`, `partner_marital_status`
  - **Constructed variable components:** `relationship_to_head` (dropped after constructing `female_head`), `first_birth_yyyymm`, and `age` (removed to avoid multicollinearity with `age_at_first_birth`)

3.1 Descriptive Statistics of Samples

Table 1 presents summary statistics of the analytical sample by female headship status. On average, female household heads are significantly older than their non-head counterparts. Although age at first birth and marriage are similar across groups, female heads exhibit significantly lower educational attainment, with a greater share holding graduate degrees. Pre-marital work experience (in months) is on the other hand significantly longer among female heads. These descriptive differences highlight systematic compositional

Table 1: Descriptive Statistics of Sample by Household Head Status

Variable	**Female Head = 1**	**Female Head = 0**
Age	49.29 (9.50)*	47.34 (10.52)
Age at First Birth	24.75 (4.46)	24.82 (4.88)
Age at First Marriage	23.73 (4.01)	23.74 (4.21)
College (share)	0.11 (0.31)	0.11 (0.32)
Junior College (share)	0.10 (0.31)	0.10 (0.30)
Graduate (share)	0.08 (0.28)*	0.14 (0.32)
PhD (share)	0.00 (0.00)	0.00 (0.00)
High School or Below (share)	0.77 (0.42)	0.78 (0.43)
Work Before Marriage (months)	410.02 (337.85)*	397.54 (343.55)

variations between the two groups, underscoring the importance of covariate adjustment in subsequent regression analyses.

Table 2 disaggregates descriptive statistics by birth cohort and female headship status. Across all cohorts, female heads tend to exhibit higher levels of educational attainment than non-heads, particularly with respect to college and graduate degree shares. Statistically significant differences emerge in several dimensions. For instance, among the youngest cohort (1985–94), female heads are significantly older and show earlier first marriage and childbirth timing. In the 1940–54 cohort, heads have significantly longer pre-marital work durations. These patterns underscore the presence of cohort-specific heterogeneity in demographic profiles and gendered labor participation, with female headship status reflecting distinct life course dynamics across generations.

Table 2: Descriptive Statistics by Birth Cohort and Female Headship

	1940–54		1955–69		1970–84		1985–94	
	Head=0	Head=1	Head=0	Head=1	Head=0	Head=1	Head=0	Head=1
Age	61.38	61.38	51.05	51.43	36.70	37.39*	25.94	26.40*
	(1.68)	(1.71)	(4.28)	(4.22)	(4.13)	(3.98)	(2.23)	(1.88)
Age at First Birth	23.14	23.43	24.76	24.61	26.04	26.08	22.90	22.52*
	(4.36)	(3.80)	(4.62)	(4.47)	(4.70)	(4.58)	(3.03)	(3.17)
First Marriage Age	22.17	22.48	23.68	23.64	24.87	24.90	22.18	21.76*
	(3.97)	(3.45)	(4.09)	(3.98)	(4.27)	(4.14)	(2.99)	(2.95)
Education: College	0.04	0.05	0.09	0.08	0.18	0.19	0.26	0.23
	(0.21)	(0.29)	(0.28)	(0.38)	(0.39)	(0.39)	(0.43)	(0.42)
Education: Graduate	0.01	0.00	0.02	0.02	0.04	0.04	0.01	0.01
	(0.09)	(0.07)	(0.13)	(0.13)	(0.19)	(0.19)	(0.11)	(0.11)
Education: High School or Below	0.90	0.90	0.80	0.81	0.63	0.59*	0.64	0.67
	(0.29)	(0.30)	(0.40)	(0.39)	(0.48)	(0.49)	(0.48)	(0.47)
Education: Junior College	0.04	0.04	0.09	0.09	0.15	0.18*	0.09	0.09
	(0.20)	(0.21)	(0.29)	(0.29)	(0.36)	(0.39)	(0.29)	(0.29)
Education: PhD	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.04)	(0.06)	(0.06)	(0.06)	(0.00)	(0.00)
Work Before Marriage (months)	350.01	371.12*	432.03	437.74	396.86	395.01	210.36	200.17*
	(339.27)	(331.12)	(343.85)	(338.72)	(347.86)	(337.95)	(200.12)	(218.21)

4 Empirical Method

To estimate the causal effect of female household headship on age at first birth, we adopt a linear regression framework augmented with the Post-Double Selection (PDS) procedure. This method combines the strengths of both LASSO (for variable selection in high-dimensional settings) and OLS (for consistent post-selection estimation), making it especially well-suited for rich, observational survey data.

Econometric Specification

The main estimating equation is specified as:

$$\text{age_at_first_birth}_i = \beta_0 + \beta_1 \cdot \text{female_head}_i + \sum_{k=2}^{K} \beta_k X_{ik} + \sum_{g=1}^{G-1} \delta_g \cdot \text{EduGroup}_{ig} + \gamma_c + \varepsilon_i$$

- $\text{age_at_first_birth}_i$: The age at first birth for individual woman i. This is our primary dependent variable.
- female_head_i: A binary treatment variable, equal to 1 if woman i is a household head, and 0 otherwise.
- β_0: The intercept term.
- β_1: The coefficient of interest, representing the estimated average difference in age at first birth between female household heads and non-household heads, holding all other covariates constant.
- X_{ik}: The k-th control variable for individual woman i, selected through the Post-Double Selection (PDS) LASSO procedure. These include demographic, employment, partner, and pre-marital history factors. All covariates are pre-outcome and screened for multicollinearity.
- EduGroup_{ig}: A set of dummy variables for education categories of woman i, where $g \in \{\text{Junior College}, \text{College}, \text{Graduate}$ and the reference group is "High School or Below". These dummies control for discrete levels of educational attainment beyond the continuous education level variable.
- δ_g: The coefficient associated with each education group dummy g, capturing average differences in first birth timing across educational strata.
- γ_c: Birth cohort fixed effects controlling for generational influences across four cohorts (1940–54, 1955–69, 1970–84, 1985–94).
- ε_i: The idiosyncratic error term.
- K: The total number of covariates selected from LASSO.
- G: Total number of education groups (5), with $G - 1$ dummies included in the regression to avoid perfect collinearity.

Motivation for Using Post-Double Selection (PDS)

The dataset used in this study derives from a nationally representative, cross-sectional survey consisting of 176 structured questions. After excluding post-treatment, identifying, and invalid variables, over one

hundred covariates remain eligible for analysis. In such high-dimensional settings, traditional OLS risks overfitting or omitted variable bias due to ad hoc manual covariate selection.

To address this, we adopt the Post-Double Selection (PDS) approach proposed by Belloni et al. (2014). This method uses LASSO regressions to identify covariates that predict either the treatment or the outcome, and includes the union of both sets in the final OLS model. By doing so, PDS ensures sufficient control for confounding while retaining a tractable model suitable for valid inference.

Unlike methods such as difference-in-differences or instrumental variables, which require panel data or exclusion restrictions, PDS is well suited for causal inference in purely cross-sectional data environments, making it particularly appropriate for our setting.

Identification Strategy

We estimate the causal effect of female household headship on fertility timing under the assumption of conditional unconfoundedness. That is, after controlling for variables selected via PDS-LASSO, along with fixed effects for both birth cohort and education group, assignment to female headship is assumed to be independent of unobserved determinants of age at first birth. This permits causal interpretation of the `female_head` coefficient as a conditional average treatment effect.

The inclusion of birth cohort and education group fixed effects accounts for systematic differences in fertility behavior across generations and educational strata—such as evolving gender norms, policy regimes, and career-family tradeoffs—thereby mitigating structural confounding.

5 Results

Variable Selection via PDS-Lasso with Thresholding

To account for potential confounding and ensure valid causal inference, we first excluded all invalid identifiers and future-leaking variables. We then employed the Post-Double Selection (PDS) Lasso procedure, which identifies variables predictive of both the treatment assignment (female headship) and the outcome (age at first birth). The full union of the first-stage (outcome-predictive) and second-stage (treatment-predictive) selections yielded 68 covariates.

To improve model parsimony while retaining predictive relevance, we applied a coefficient magnitude threshold of $|\beta| \geq 0.02$ and retained 31 covariates with non-negligible effects. As shown in Table 3, the thresholded model achieves comparable fit to the full PDS-Lasso model with a reduced number of variables, lowering the Bayesian Information Criterion (BIC) from 38365 to 38077. This filtered model is used in our main specification.

Table 3: Model Fit Comparison Across OLS and PDS-Lasso Variants

Model	**Vars**	**Adj-R^2**	**AIC**	**BIC**	**RMSE (train)**	**RMSE (test)**
OLS	103	0.8591	37895.87	38534.87	1.6913	1.8179
Lasso_Full	68	0.8591	37870.14	38365.55	1.6925	1.8169
Lasso_0.02	**32**	**0.8591**	**37840.46**	**38077.39**	**1.6962**	**1.8176**
Lasso_0.01	36	0.8590	37845.70	38111.35	1.6960	1.8186

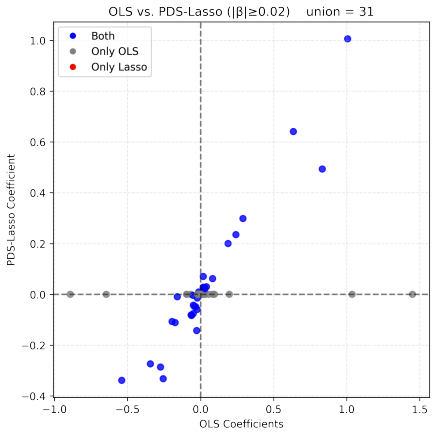

Figure 1: OLS vs. PDS-Lasso

Post–Lasso covariates after the $|\beta| \geq 0.02$ threshold ($k = 31$)

- **Demographics and household composition**
  `num_adults_in_household`, `marital_status`, `foreign_spouse`
- **Own education**
  `high_school_or_below`, `junior_college`, `graduate`, `phd`, `schooling_status`
- **Fertility preferences**
  `firstmarry_age`, `spouse_ideal_boy`, `spouse_ideal_girl`
- **Employment status and mobility**
  `changejob_aftermarry`, `backtowork_aftermarry_type`, `backtowork_aftermarry_reason`, `work_before_marriage_type`
- **Job-search behaviour and reasons for non-work**
  `jobsearch_status`, `jobsearch_fail_reason_1`, `nojob_reason`, `quit_marry_reason`, `reason_left_last_job`
- **Partner's attributes and labour participation**
  `partner_education_level`, `partner_schooling_status`, `partner_full_or_part_time_pref`, `partner_paid_or_unpaid_work`, `partner_reason_left_last_job`, `partner_reason_not_working`, `partner_want_more_work_hours`, `partner_job_search_method_3`, `partner_job_search_method_4`, `partner_job_search_method_6`, `partner_respondent_self`

Note: Several covariates in the model are categorical indicators derived from multi-option survey questions. For instance, `jobsearch_fail_reason_1` captures the main reason for failed job search, with values including: "no suitable jobs available," "lack of required skills," or "care responsibilities." Likewise, `nojob_reason` indicates why a respondent is currently not working, covering responses such as "housework," "health issues," or "education." The variable `quit_marry_reason` records the reason for leaving employment at the time of marriage, such as "spouse requested," "family expectation," or "personal preference." On the partner side, job search method dummies like `partner_job_search_method_3` or `partner_job_search_method_6` denote whether the partner searched via government referral, private agency, or internet. Variables such as `partner_reason_left_last_job` and `partner_reason_not_working` describe the partner's employment trajectory, including retirement, layoffs, or household obligations. These dummy variables are constructed to capture latent labor market frictions and life-course decisions relevant to both spouses' fertility timing.

Table 4: OLS Estimates for Age at First Birth: Robustness Checks

	Full Sample	**Add Cohort FE**	**Married Only**
Female Headship			
female_head	-0.0990	-0.0762	-0.0734
	(0.042)	(0.041)	(0.041)
Fertility Preferences			
firstmarry_age	1.0036***	1.0059***	1.0018***
	(0.005)	(0.005)	(0.005)
spouse_ideal_boy	0.3802***	0.2398	0.3850***
	(0.097)	(0.095)	(0.094)
spouse_ideal_girl	0.2984***	0.2984***	0.2984***
	(0.101)	(0.101)	(0.101)
Education (Own)			
high_school_or_below	-0.2454***	-0.2604***	-0.2086***
	(0.060)	(0.065)	(0.065)
graduate	0.1646	0.1999	0.1726
	(0.115)	(0.132)	(0.132)
phd	0.2980	0.6409*	0.8734**
	(0.102)	(0.353)	(0.342)
junior_college	0.0495	0.0476	0.0495
	(0.075)	(0.076)	(0.081)
schooling_status	0.0420	0.0147	0.0147
	(0.038)	(0.036)	(0.066)
Demographics & Household			
marital_status	-0.3782***	-0.3323***	–
	(0.080)	(0.079)	–
foreign_spouse	-0.3076***	-0.2738***	-0.4783***
	(0.066)	(0.065)	(0.102)
num_adults_in_household	0.0528***	0.0476***	0.0459***
	(0.012)	(0.012)	(0.013)
Partner Attributes			
partner_education_level	0.0921***	0.0826***	0.0415***
	(0.012)	(0.012)	(0.012)
partner_schooling_status	-0.0759*	-0.0862*	-0.0862*
	(0.045)	(0.049)	(0.049)
partner_want_more_work_hours	-0.1232	-0.1113	-0.1103
	(0.075)	(0.075)	(0.075)
partner_paid_or_unpaid_work	-0.0801***	-0.0892***	-0.0290
	(0.013)	(0.013)	(0.013)

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	Full Sample	Add Cohort FE	Married Only
Employment Status & Job Search			
employment_status	–	–	0.0211
	–	–	(0.027)
jobsearch_status	0.0116*	0.0076	0.0087
	(0.006)	(0.006)	(0.010)
jobsearch_fail_reason_1	0.0654	0.0615	0.0903
	(0.053)	(0.060)	(0.053)
Partner Job Search			
partner_full_or_part_time_pref	0.1024*	0.1071*	0.0911
	(0.047)	(0.047)	(0.081)
partner_job_search_method_3	0.5458**	0.4933**	1.5575
	(0.187)	(0.188)	(1.081)
partner_job_search_method_4	-0.3441	-0.3387	1.6923
	(0.152)	(0.531)	(0.549)
partner_job_search_method_6	0.0710	0.0723	0.2970
	(0.154)	(0.154)	(0.185)
Marriage History			
backtowork_aftermarry_reason	0.0229	0.0250	0.0214
	(0.024)	(0.024)	(0.020)
backtowork_aftermarry_type	0.1436***	0.1428***	–
	(0.043)	(0.043)	–
changejob_aftermarry	0.0346	0.0281	–
	(0.027)	(0.027)	–
work_before_marriage_type	-0.0706***	-0.0597***	-0.0597***
	(0.020)	(0.024)	(0.024)

Notes: Estimates are based on Post-Double Selection Lasso covariates with $|\beta| \geq 0.02$ threshold. Robust standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Across all models, `firstmarry_age` consistently shows a strong positive effect on age at first birth. Respondents with foreign spouses tend to have earlier childbirth, especially within the married-only sample. The variable `female_head` is associated with lower age at first birth, though the magnitude shrinks slightly after including birth cohort fixed effects. Educational variables such as `phd` and `high_school_or_below` remain influential in the expected directions.

The cohort fixed effect specification does not significantly change most estimates, suggesting stability across generations. The model restricted to married women excludes `marital_status` by construction and yields similar coefficient signs and magnitudes, supporting robustness to marital selection.

5.1 Main Estimates

Table 4 presents the OLS estimates for age at first birth under three specifications: (1) full sample with all covariates selected via PDS-Lasso, (2) controlling additionally for cohort fixed effects, and (3) restricting the sample to currently married respondents. Across all models, the treatment variable `female_head` is consistently associated with a **lower** age at first birth. In our preferred specification (Column 2), the

estimated effect is -0.0762 with a robust standard error of 0.041, indicating that female household heads tend to have their first child slightly earlier than comparable non-head women after adjusting for observed confounders and generational shifts.

5.2 Interpretation of Key Covariates

The coefficient on `firstmarry_age` remains stable and large (≈ 1.00) across specifications, confirming its strong predictive power over fertility timing. Educational attainment also displays intuitive gradients: respondents with a PhD tend to delay childbirth, while those with high school education or below tend to have children earlier. Partner attributes such as education and labor force engagement also show significant associations.

5.3 Robustness Across Subsamples

Model (3) restricts the analysis to married respondents only. The treatment effect remains negative and significant, albeit slightly attenuated. This consistency suggests that the estimated effect is not solely driven by differences in marital status or household composition.

5.4 Model Selection Justification

We apply a PDS-Lasso covariate selection with a coefficient magnitude threshold of $|\beta| \geq 0.02$, resulting in a parsimonious model with 31 covariates. This approach balances model complexity and interpretability, reducing the BIC from 38365 (full model) to 38077. Importantly, we explicitly controlled for both birth cohort and education group to address secular shifts and nonlinear educational gradients in fertility behavior.

6 Discussion and Conclusion

This paper explores the relationship between female household headship and age at first birth, using nationally representative survey data from Taiwan. Leveraging the Post-Double Selection Lasso method, we control for a rich set of covariates while minimizing researcher discretion. Our findings suggest that women who are household heads tend to have their first child **slightly earlier** than non-heads, even after controlling for educational attainment, cohort trends, and partner characteristics.

This result is somewhat counterintuitive, as one might expect greater autonomy to correlate with fertility postponement. However, it is possible that female headship in the Taiwanese context may reflect earlier life disruptions (e.g., divorce or widowhood) rather than proactive autonomy, consistent with concerns about reverse causality discussed earlier.

Our analysis is robust across model specifications and sample restrictions. Still, several limitations remain. The cross-sectional nature of the data restricts our ability to establish clear temporal ordering between treatment and outcome. Unobserved variables—such as fertility intentions or health constraints—may still bias the estimates. Future research using panel data or quasi-experimental variation (e.g., policy changes affecting household structure) would offer stronger causal identification.

Overall, this study contributes to the understanding of how intra-household roles relate to reproductive behavior in East Asia, where both educational attainment and delayed fertility are rising. By quantifying the link between household authority and fertility timing, we add nuance to existing debates on autonomy and reproductive choice.